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Solving strategic management challenges with Design Thinking: examining the suitability and effectiveness as a management method

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Abstract

Design Thinking (DT) is widely recognized for its role in product and service innovation. However, its application as a management method, where executives employ to address their own management problems, remains underexplored. Previous research on DT in management discourse has mainly focused on its impact on organizations when infused at scale, analyzing organizational innovativeness, innovation culture, or product innovation. Given that DT is primarily a human-centric approach, it may seem incompatible with addressing abstract strategic management problems. This raises questions about the DT's applicability and effectiveness in addressing managerial challenges. This paper presents an interview-based qualitative study evaluating DT's potential for solving wicked management problems. The study illuminates the dilemmas and peculiarities associated with applying DT in a management context. This paper offers novel insights into DT's potential as a transformative tool in strategic management, bridging the gap between traditional management methods and design-led approaches to strategy formulation and implementation.

Keywords: Design Thinking, Managerial design thinking, Problem solving, Strategic management, Management method

Introduction

Contemporary organizations face increasingly complex strategic management challenges that defy conventional problem-solving approaches. The rapid pace of technological change, market volatility, and evolving stakeholder expectations create what Camillus (2008) terms "wicked problems"—interconnected and ambiguous challenges that conventional analytical methods frequently prove inadequate to address. This strategic decision-making crisis is evidenced by the declining life span of S&P 500 companies, where average tenure has dropped from 32 to 21 years, with projections indicating nearly half will be replaced in the next decade (Clark, 2021). This organizational mortality rate underscores the urgent need for more effective approaches to strategic management challenges.

The observed decrease in organizational lifespan stems partly from the limitations of descriptive management approaches, which inadequately address emerging variables in

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strategic decision-making (Anthony et al., 2016). Strategic decisions predominantly rely on descriptive, historical, and market data, often neglecting emergent variables (Carlopio, 2009). Pries-Heje and Baskerville (2010) argue that descriptive management styles are insufficient for solving present management challenges since they reflect the organizational past but do not necessarily shape the future.

DT emerges as a potential response to these strategic management challenges. Its human-centered and systems-oriented perspective, proven effective in innovation contexts, offers a powerful yet underutilized framework for addressing high-level strategic challenges beyond traditional method's scope. Generative and prescriptive methods that combine analytical and design capabilities prove more effective in solving contemporary management challenges (Powell, 2008). These may lead to novel solutions that are better suited to current management challenges (Pries-Heje & Baskerville, 2010; Boland et al., 2008; Martin, 2009; Liedtka, 2018).

Design Thinking provides a generative approach underpinned by a design attitude to problem-solving that facilitates decision-making and leadership in modern organizations (Matthews & Wrigley, 2017; Zheng, 2018). It addresses what are commonly characterized as wicked problems—complex issues framed by comprehensive problem statements, known as design challenges. These design challenges typically encompass narratives that reflect the issue's complexity, the users affected, contextual details, and the constraints or conditions that must be overcome. The wickedness of these problems arises from overwhelming complexity, wherein pure rationality and analytical approaches often prove inadequate, positioning DT as a viable alternative (Bauer & Eagan, 2008). The precise articulation of a design challenge remains critical, as phraseology significantly influences solution development trajectories (Lindberg et al., 2010).

While DT shows promise for strategic challenges, its problem-solving capabilities offer an approach to address strategic business challenges that are genuinely wicked and abstract rather than human-centric (Baran, 2017; Dunne & Martin, 2006; Glen et al., 2014; Graf, 2021; You, 2022). Specifically, strategic management challenges are framed within the scope of DT but without the typical immediacy of a consumer or user focus. The predominant human-centric focus of DT has been critiqued for potentially overlooking strategic imperatives in managerial contexts (Elsbach & Stigliani, 2018). Shapira et al. (2017) argue that Design Thinking may require a comprehensive systems perspective. Recent research also suggests that DT and strategic management converge in their approach to unraveling complex and interrelated systemic challenges through the lens of systems theory, which is pivotal for fostering informed decisions and innovative solutions (Gianelloni & Goldstein, 2020). The inherent wickedness of strategic management problems necessitates a holistic view of the business situation (Camillus, 2008). This introduces a methodological tension because strategic business challenges are more generic and abstract than human-centric design challenges (Björklund et al., 2018). Applying DT with a primarily human-centric approach to an abstract management problem can lead to a dilemma and potentially limit its effectiveness as a management method.

To address these methodological challenges, researchers have begun exploring managerial DT as a distinct approach devoted to solving complex management problems and fostering the creation of novel opportunities. This adaptation harnesses both design and

management tools within the DT framework to develop generative and inventive solutions for management. The design challenges encountered in managerial DT are typically broad in nature, encompassing considerably higher uncertainty than traditional DT engagements. Given that managerial tools are pivotal in facilitating effective decision-making, problem-solving, and strategy development (Berhisha Qehaja et al., 2017), managerial DT can support the strategic decision-making process, generate unconventional management solutions, and aid in their implementation (Bender-Salazar, 2023; Rösch et al., 2023; Baran, 2017; Christensen & Nielsen, 2019; Elsbach & Stigliani, 2018; Graf, 2021; Micheli et al., 2019; Liedtka, 2018). The discussion in this paper situates DT for solving abstract managerial problems and the related avenue of DT as 'a process of exploration and creative strategies' and hence as a management practice (Dorst, 2011).

Design Thinking has been widely recognized for its efficacy in human-centric design challenges, particularly those related to product and service innovation, thereby contributing to organizational innovativeness and culture (Knight et al., 2020; Elsbach & Stigliani, 2018; Daymond & Knight, 2023; Liedtka, 2011; Carlgren et al., 2016; Brown, 2008; Plattner, 2009). Other scholars have examined the design activity perspective of Design Thinking and non-designer innovation capacities, focusing on the differences between Design Thinking and designerly thinking (Seidel & Fixon, 2013; Johansson-Sköldberg et al., 2013).

Despite these advances in understanding, a significant gap persists concerning the concrete application of DT for solving complex, multi-layered management problems (Liedtka & Kaplan, 2019; Graf, 2021; Bühring & Liedtka, 2018; Liedtka, 2018). Although the management research literature acknowledges the possibility of developing management solutions through design, it does not elaborate on the details of this practice (Belyaeva et al., 2020; Dunne & Martin, 2006; Knight et al., 2020; Liedtka & Ogilvie, 2011; Martin, 2009; Schmiedgen et al., 2016). Consequently, research predominantly hypothesizes DT's potential for strategic management yet lacks concrete examples or in-depth discussions of its application with managers and executives. Details that provide sufficient guidance in applying DT for novices, such as managers and executives, are lacking (Laursen & Haase, 2019). Documented instances of DT engagements with managerial challenges that influence a company's strategic direction are notably scarce (Diderich, 2020). Moreover, comprehensive research discussing the suitability of various design challenge types, especially those that encompass truly wicked, multi-layered, managerial, and micro- or macro-economic design challenges, is limited. Comparative studies measuring DT's efficacy against other problem-solving and innovation methods are likewise absent. The evolving complexity and ambiguity of today's strategic business challenges further underscore the urgent need for research bridging these gaps, thereby examining DT's full range of problem-solving capabilities.

Understanding the applicability and nuances of employing DT for management problems is crucial for leaders seeking novel ways of solving management problems. Additionally, understanding how Design Thinking can be adapted and then applied by executives to address these 'wicked' problems could provide critical insights into effective strategic management methodologies. An interview-based qualitative study with experienced DT practitioners and facilitators, who have worked with and observed leaders and executives in such engagements, can address this gap, offering details on the

practical application of DT in strategic management engagements that approach complex, truly wicked management problems. Addressing this gap requires insights into DT's applicability, effectiveness, and implementation strategies for managerial design challenges. Such insights can also guide the evolution of classic DT into a managerial DT method that remains faithful to its foundational principles.

A wide array of management challenges can be addressed through managerial DT, including sales and go-to-market strategies, strategic portfolio management, performance management, financial planning, intrapreneurship, resource and talent management, process and operations management, and business model innovation (Bühring & Liedtka, 2018; Christensen & Nielsen, 2019; Elsbach & Stigliani, 2018; Hehn & Übernickel, 2018; Liedtka & Kaplan, 2019, You, 2022). These types of challenges typically necessitate allocating only one or two days from executives' and leaders' agendas.

This study addresses these research gaps through a rigorous empirical investigation employing Design Science Research methodology. This methodological approach specifically tackles the identified gaps in three ways: examining DT's application in strategic contexts, assessing DT's implementation challenges and success factors, and developing evidence-based guidelines for adapting DT to strategic management challenges. This study therefore contributes to filling this research gap by offering essential empirical insights into DT's applicability for tackling complex managerial issues, thereby expanding its scope as a transformative management method beyond its well-documented role in product innovation.

To address these critical research gaps, this study poses the following research question: how can Design Thinking be effectively applied by executives to solve wicked problems in strategic management, and what are the key challenges and success factors in its implementation?

The research goal is to critically evaluate DT's applicability and effectiveness in strategic management, given its predominantly human-centric approach to problem-solving. Specifically, this study aims to investigate how DT can shape decision-making processes and strategy formulation by conducting an in-depth analysis of its implementation, effectiveness, and challenges in real-world business scenarios. Ultimately, the study seeks to offer valuable insights into the potential of Design Thinking as a transformative tool for modern management challenges.

First, we examine the extent to which DT is suitable for solving wicked management problems. Second, we assess the acceptance of DT as a management method among its target audience. Finally, we identify the characteristics of managerial DT and suggest mitigation strategies to resolve potential dilemmas while adhering to Design Thinking principles.

Theoretical background

Design Thinking, recognized for its problem-solving and innovation capabilities, is primarily employed for developing innovative products and services (Brown, 2008; Martin, 2009; Plattner et al., 2009; Liedtka, 2014). As Verganti et al. (2020) assert, problem-solving lies at the core of the innovation process, with the associated decision-making practice constituting what can be termed as designing (Verganti et al., 2020).

A typical feature of DT is its human-centric focus, which includes the ability to address "wicked" problems, separate the problem from the solution space, employ exploration and ideation tools, create artifacts, and use abductive reasoning (Johansson-Sköldberg et al., 2013; Laursen & Haase, 2019; Micheli et al., 2019).

Laursen and Haase (2019) assert that "Design Thinking aims to export designers' ways of thinking to non-designers, with the most straightforward way of doing so by using tools and techniques." DT seeks to equip the non-designer with a "cookbook approach"—a guided, phased process supplemented with various tool options in each phase. Although this cookbook approach may help overcome initial obstacles to using design principles for problem-solving, it also poses a risk of misapplying certain techniques to a problem (Laursen & Haase, 2019).

Because DT is both broad and flexible, it offers a dynamic methodological value proposition that varies on its application, context, and problem granularity. It appears particularly well-suited for wicked and complex problems because it allows extensive exploration of both the problem and solution space. Solution design or development commences only after establishing an adequate understanding of the problem and gathering the necessary insights (Carlgren et al., 2014).

Design Thinking is positioned not only as an innovation method grounded in design principles, but also as a broader problem-solving approach especially suited to addressing these "wicked problems" (Camillus, 2008; Kimbell, 2011; Plattner et al., 2009). The complexity of such problems varies significantly between individual needs and organizational "wicked business problems" (Borja de Mozota, 2010). The wickedness of strategic management challenges is likely even more complex due to micro- and macro-economic influential variables, further complicating the systematic discovery of cause and effect (Graf, 2021).

Design Thinking and strategic management both require decision-making on highly complex, non-linear, and wicked problems in the context of a larger system. Integral to both domains is the imperative of understanding systemic and behavioral complexities and their interrelationships. Systems theory is central to this understanding and supports better problem exploration, informed decision-making, and innovative solution development (Gianelloni & Goldstein, 2020). It provides scientifically grounded frameworks for understanding and managing these complexities, even when they are multifaceted and apparently unrelated. Currently, Systems Thinking, as an application of systems theory, is gaining prominence for addressing complex challenges in both classic DT and strategic management. It advocates a perspective that looks beyond the immediate problem to holistically understand the broader system, context, and dynamics (Grewatsch, 2023; Kerguenne et al., 2023; Mugadza & Marcus, 2019; Forsythe, 2020).

Design cognition theories emphasize the mental processes involved in designing solutions, showcasing how DT can transform managerial thinking and encourage innovative solutions to strategic issues (Hay et al., 2017). Practice theories examine the routines that support strategic action, demonstrating how DT's collaborative methods empower managers to tackle strategic challenges with a flexible and adaptive approach (Romme, 2016). Additionally, DT is positioned within cognitive frameworks as a sensemaking activity that enhances organizational sensibility through imagination and adaptive thinking, thereby supporting continuous learning in complex environments (Rylander Eklund

et al., 2021). From a practice theory perspective, DT is also framed as a dynamic capability that relies on individual-level skills and routines—such as iterative prototyping and user-centered research—to build both practical adaptability and cognitive flexibility, enabling organizations to sense, seize, and shape opportunities in rapidly changing contexts (Magistretti et al., 2021). This synthesis of cognitive and practice-based theories positions DT as a versatile and potent tool within strategic management, offering a structured yet adaptive approach to transformative, complex, ever-evolving management problems (Kerguenne et al., 2023).

Solving management problems with a design attitude has been discussed as an alternative approach alongside DT's growing popularity (Çeviker-Çınar et al., 2017; Graf, 2021; Liedtka, 2000). Dunne and Martin (2006) argue that "approaching management problems as designers approach design problems" offers significant advantages for today's managers and executives. Carr et al. (2010)further contend that "Design Thinking applies to any business, whether or not it is product-related" and that it is a way of "… finding innovative solutions to all categories of business challenges."

Recent bibliometric investigations indicate a growing scholarly interest in DT within management discourse, reflecting steady growth over recent years. Studies show that the research interest in DT's problem-solving capability surpasses its innovation capability and that DT is expanding to a wide array of application areas in business (Bhandari, 2023; Khurana & Gupta, 2023).

Lately, a trend has emerged to combine DT with other managerial methods, including Lean Startup and Business Model Innovation. The latter has become increasingly popular among companies seeking to generate additional revenue or transform their businesses (Thoring & Müller, 2023; Bonakdar & Gassmann, 2016; You, 2022). DT is viewed as an accelerator or catalyst for creating new strategic opportunities and differentiating from competitors, potentially leading to market leadership and, in some instances, also defining markets (Abrell, 2016; Christensen & Nielsen, 2019; Liedtka, 2018; Liedtka & Kaplan, 2019). Recent case studies in predominantly management literature discuss examples of effectively solving managerial design challenges with DT. These include the software, packaging, and fast-moving consumer goods (FMCG) industries, which showcase concrete applications of DT in strategizing (Efeoglu & Møller, 2023; You, 2022; Diderich, 2020; Jaskyte & Liedtka, 2022; Graf, 2021).

In their 2023 case study, Vendraminelli and colleagues examine an Italian luxury eyewear company, illustrating the practical application of DT at the executive level in designing and implementing a business transformation strategy. The company addressed a strategic challenge to*'navigate the complexities and uncertainties inherent in digitally enabled business transformation by aligning with strategic objectives and improving operational efficiencies for higher consumer engagement.'* The shift from a traditional best practices strategy to an opportunity-based one enabled the company to leverage new technologies for greater efficiency and value creation. This pivot created new business opportunities and led to a significant reconfiguration of the operating model and production processes (Vendraminelli et al., 2023). Additionally, Efeoglu and Møller (2023) provide an example of establishing an intelligent packaging strategy intended to support future digital business models, thereby optimizing shipping and delivery processes to facilitate comprehensive consumer data collection.

Past literature suggests that the design domain often demands specific skills and cognition from a designer to safeguard the profession. Whether nondesign professionals such as managers can adopt this mindset remains in question. Common design capabilities, such as contextualizing information and visualization, can be adopted by managers to enhance their inductive and deductive thinking by adding abductive reasoning (Brown, 2008; Johansson-Sköldberg et al., 2013; Liedtka, 2015; Micheli et al., 2019).

In current management discourse, Design Thinking can be debated as both a source and a resource (Brandes et al., 2014; Dell'Era et al., 2020; Liedtka & Kaplan, 2019; Coco et al., 2020). As a *source*, DT entails applying its principles to a specific challenge to generate management solutions for strategic business problems (Cagnin, 2018; Knight et al., 2020; Micheli et al., 2019). As a *resource*, DT refers to its systematic adoption aimed at cultivating innovation across the organization and securing strategic leadership in the market or industry (Brenner et al., 2016; Brown, 2008; Martin, 2009; Camillus, 2008; Badke-Schaub et al., 2010; Verganti, 2008; Kimbell, 2011; Carlgren et al., 2016; Dell'Era et al., 2020). Stevens, Moultrie, and Crilly (2008) argue for a clear distinction between DT as a management practice (*source*) and DT as a strategy (*resource*): "The management of an integrated design approach to implement a strategy is a fundamentally different activity from the use of design methods to inform a strategy."

Contemporary research indicates that companies do not fully capitalize on DT's capabilities, whether as a source or a resource. A gap exists between the DT knowledge and skills learned and their application within corporate environments. Time constraints, combined with issues of empowerment, have been identified as significant barriers. The discrepancy between acquiring DT knowledge and its subsequent practical application—often referred to as the 'practice gap'—is anticipated to be even more pronounced at higher management levels, given the limited time availability (Mayer et al., 2023).

DT as a management practice holds considerable potential because strategy design inherently depends on interactivity and dialogue with key internal stakeholders. Engagement from executives and leadership is vital for effectively addressing strategic management challenges. Design Thinking aligns with these principles, making it a powerful tool for strategy formulation. Solving strategic management problems, therefore, relies on collaborative efforts, shared understanding, and collective intelligence, as the complexity of such challenges can exceed the cognitive capabilities of even the most astute leaders. This is particularly true as ambiguity and unpredictability intensify, necessitating the integration of diverse insights from a broad spectrum of stakeholders and experts (Andersen, 2023). DT synthesizes historical, current, and future insights, employing both descriptive and prescriptive approaches to strategizing, characterized by progressive abductive reasoning, a systemic viewpoint, a value-centric approach, tolerance for failure, and adaptability in solution design. Accordingly, DT enhances strategic thinking by fostering creativity, intuition, reflection, and openness to change when addressing strategic management challenges (Diderich, 2020; Graf, 2021; Simeone et al., 2023). Furthermore, DT's emphasis on collaboration and stakeholder involvement facilitates stronger cohesion between strategy formulation and execution, as well as broader stakeholder alignment (Bühring & Bishop, 2020).

As a result, design-inspired leaders are increasingly discovering DT's problem-solving and innovation capabilities for management challenges (Boland et al., 2008; Koomans

& Hilders, 2016; Sarasvathy et al., 2008). Recent literature suggests that combining DT with traditional analytic management tools is a practical approach to addressing contemporary organizational challenges (Diderich, 2020; Knight et al., 2020; Liedtka & Kaplan, 2019).

This study elaborates on Design Thinking as a practice for management (*source)* and its design-driven capabilities for addressing systemic business challenges, with particular emphasis on executive audiences.

Method

In this qualitative study, Design Science Research (DSR) served as a guiding framework for an interview-based exploration of DT's suitability for tackling managerial challenges, particularly within limited time constraints. Qualitative methods are especially suitable for investigating emerging, context-dependent phenomena such as managerial DT, where rigid quantitative metrics may fail to capture the richness and variability of responses (Denzin & Lincoln, 2005). The DSR framework, which emphasizes solving real-world problems while contributing to the knowledge base, is well-suited for this study (Hevner et al., 2004). It provides a design-oriented and problem-focused approach that structures the investigation around the adaptation and evaluation of DT as a methodological artifact.

DSR informs the research question and the data collection process, ensures alignment with theoretical constructs, and enables the refinement of the DT method based on both theoretical and practical insights. This approach allowed for a thorough investigation into DT's methodological suitability for managerial challenges, evaluating its applicability through expert interviews and helping bridge the gap between the practical application of DT in strategic management and the theoretical discourse.

This study aimed to understand the distinction between managerial DT and traditional DT, and to determine whether it must be applied differently or adapted for abstract managerial challenges. The insights focus on DT methodology, techniques, management decision behavior, and solution commitment, examining how managerial DT engagements are adopted as a management method for strategic decisions.

Interviews are an appropriate data collection method when the researcher has sufficient professional, academic, and practical knowledge to effectively lead discussions with peers. The researcher's ability to conduct credible discussions using an effective interviewing approach enhances the study's validity and reliability (Kvale, 1996; Kvale, 2008; Patton, 2005). The interview questions were carefully designed to solicit responses relevant to the research objectives. Drawing on established research in strategic management and Design Thinking (e.g., Carlgren et al., 2016; Liedtka, 2018), questions focused on decision-making behavior, solution commitment, and DT's perceived effectiveness in solving complex managerial challenges. By aligning interview questions with these focal points, the study obtains essential insights into how Design Thinking functions as a management tool.

The selected interviewees were innovation managers and business consultants with extensive DT skills, each having facilitated more than 50 management-related DT engagements. The sampling strategy prioritized DT expertise, prior experience with strategic management engagements, number of engagements, and diversity of strategic

management challenges solved through DT. These engagements often lead to business transformations and are undertaken by managers and executives who define new strategies or aim to reinvent their businesses. The theoretical sampling strategy with expert sampling used in the study enabled the selection of 10 DT experts who were actively involved in solving strategic management challenges using DT.

Sampling began by identifying expert practitioners from an organizational pool of approximately 180 DT experts and facilitators. The list was narrowed to those who support both internal projects and external clients, then further stratified to include client-facing experts dealing with managerial design challenges. From this list, those with the greatest engagement experience were selected. The experience criteria for selection ranged from 1 to 10 engagements, then up to 20, up to 50, and 50 and above. This threshold was established to ensure that facilitators possess the depth of insight needed to evaluate DT's strategic applications and limitations. This sampling strategy aimed to form an interview group of DT experts and facilitators with the most extensive engagement experience, thus equipping the study with a robust foundation of insights.

Data saturation from the interviews facilitated the derivation of synthesized insights and the formulation of conclusions on DT's suitability as a modern management method. The semi-structured interviews with DT experts were designed to obtain their perspectives and experiences when solving strategic management problems using DT. A series of open-ended and closed questions were asked to achieve this objective. Closed questions were used to facilitate the interview process and draw out interview participants' opinions on the suitability of DT as a managerial tool. The open-ended questions focused on DT and management-specific themes and were designed to generate in-depth responses. Ad hoc and selective questions were also employed to gather further insights and clarify specific points. Participants were also asked to offer recommendations for improving DT's application in managerial contexts. Inverse questions were used to obtain a more comprehensive understanding of the topic and to assess objectivity by exploring the potential drawbacks of managerial DT.

Data collection

For this study, data were collected through semi-structured interviews that drew on effective questions based on previous qualitative and quantitative research on Design Thinking. To ensure reliability and validity, the study not only selected highly experienced practitioners but also employed strategies such as triangulation, member-checking, question synthesis from previous research, and an iterative question refinement process. Pilot interviews were conducted to assess the clarity and relevance of the questions, leading to refinements that enhanced data quality and rigor.

The questions addressed organizational, human, and mindset factors, as well as methodology, technique, duration, business challenge, solution, and implementation aspects of strategic management engagements. Because all the interviewees had extensive knowledge and experience in management advisory using DT, and continue to work in the same field, they were able to provide deep insights.

Each interview lasted between 90 to 180 min and was conducted face-to-face or via teleconferencing. All interviews were recorded and transcribed. This qualitative study

thereby elucidates the "why", "how", and "what" behind management solution generation via DT. Table 1 presents example interview questions.

Data analysis

Data from the open-ended interview questions were coded and condensed into meaningful information and insights. The analysis involved both open and axial coding. A comprehensive category system covering all aspects of the interviews was developed to guide the coding process. Coding can be conducted by focusing on specific insights or filtering data (Flick, 1995; Mayring, 2000). The coded insights were then validated against established categories such as definitions, insights, patterns, and behavior (Burnard, 1991). Because the data collected are context-specific and sensitive to the DT problem and solution space (Priest et al., 2002), data conversion, semantic coding, and interpretation were conducted in the context of each DT-space.

The transcribed responses capture the full breadth of the interviewees' perspectives. For each question, the responses were all listed, and key insights relevant to the managerial DT method and the research question were emphasized. A systematic open coding process was then employed to identify relevant statements and key expressions. This phase focused on organizing the qualitative data into central themes such as key messages, issues, concerns, facts, roles, and behaviors.

Subsequently, these insights were clustered semantically, ensuring that coding remained context-sensitive, particularly regarding DT's distinct phases in the problem and solution spaces. This coding is sensitive to the nuances of each phase within their respective problem and solution space, requiring a high degree of manual coding to minimize potential semantic errors that might emerge due to automation and software use. Nonetheless, software was deliberately employed to support the coding for mapping and further codification. The codification process yielded several major themes: management challenges, DT methodology, application, results, and managerial behavior. The methodological aspects were further subdivided into constructional and meta-methodological dimensions.

Following open coding, the data underwent axial coding. In this phase, the insights were reorganized, and robust connections were established between themes, categories, and subcategories, thereby illuminating context, assumptions, and consequences. This recontextualization was critical to interpreting the data, synthesizing findings

Table 1 Sample interview questions

Methodological magnitude	
What corrupts a managerial DT engagement?	**fh**
How comfortable are managers and executives when exposed to DT for the first time?	
Managerial magnitude	
How do managers and executive participants commit to the realization of the managerial DT results?	
What are the advantages/disadvantages of managerial design thinking when compared to other management methods?	
Organizational magnitude	
For which type of business or management topic is managerial DT best applicable for?	
What constitutes good managerial DT results, in your opinion?	

Table 2 Data structure for axial coding

Methodological magnitude	Managerial magnitude	Organizational magnitude
DT-spaces and -phases	Business problem statement comprehension	Organizational complexity
Techniques	collaboration and contribution	Organizational capability
Divergence and convergence	Decision-making and democracy of opinions	Strategy to execution
Duration and time management	Analytic vs. creative mindset	Project and program management
Method acceptance	Management solution validation	Stakeholder management
Iteration	Commitment to management solutions	Transformation and change management
Facilitation	Hidden management agendas	Efficiency and effectiveness
Adjacent or complementary methods	Work delegation	

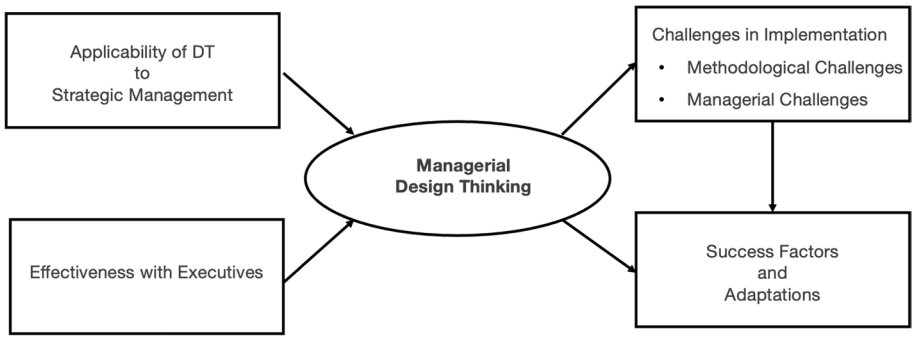


Fig. 1 Conceptual framework for applying DT in strategic management

with existing theory, and articulating conclusions and recommendations. Axial coding revealed the interdependencies, emphasized the benefits and drawbacks of managerial DT, and identified critical success factors and their defining characteristics.

The themes and categories identified through open and axial coding serve as a framework for systematically answering the research question. The new insights about methodological peculiarities and 'DT as a management practice' help fill gaps in the existing literature. Table 2 presents the axial coding categories. The data analysis process was systematic and rigorous to ensure both reliability and validity of the study's findings.

Results

This chapter presents the study's findings, organized thematically to address the research question: "How can Design Thinking be effectively applied by executives to solve wicked problems in strategic management, and what are the key challenges and success factors in its implementation?" The thematic organization of the findings offers a comprehensive understanding of DT's applicability to strategic management, its effectiveness when employed with executives, the challenges encountered during implementation, the key success factors, and the adaptations necessary for managerial contexts.

The following diagram illustrates these key components and their interrelationships arising from the study, bringing them together in a conceptual framework (Fig. 1).

Applicability of design thinking to strategic management

Our research demonstrates that DT is particularly effective for resolving complex, wicked problems in strategic management. Although DT is traditionally associated with human-centric product and service innovations (Brown, 2008; Plattner et al., 2009; Liedtka, 2014), its principles can be adapted to address higher-level, systemic challenges faced by executives.

Systemic view of management problems

A critical finding is the need for a systemic perspective when applying DT to managerial challenges. All DT engagements begin with an initial phase aimed at establishing a common understanding of the challenge, which is essential given executives' diverse yet complementary responsibilities. This phase functions as a "calibration phase" for comprehending the scope and direction of the management challenge.

> *"Common understanding is the fundamental basis for whatever comes after in understanding the problem statement." (Interviewee #3)*

Managerial design challenges differ from traditional DT challenges due to their abstract, systemic nature. They are rarely persona-specific and tend to be broadly defined and high-level, often lacking a direct consumer or user focus (Björklund et al., 2018). Despite this abstraction, DT principles—including separation of problem and solution spaces, abductive reasoning, and collaborative exploration—remain applicable. DT generally anticipates a well-understood scope and engagement goal (Grots & Pratschke, 2009; Vetterli et al., 2012).

The study indicates that an ill-defined managerial design challenge can shift the engagement from its original intention, whereas a well-defined challenge can lead to faster consensus and more successful outcomes. This finding aligns with Lindberg et al. (2010), who highlight the critical importance of clear articulation in DT engagements.

> *"There is a big difference between managerial DT and classic DT in its nature due to time constraints, involvement of end-users, and design challenge type." (Interviewee #2)*

Effectiveness of Design Thinking with executives

DT's effectiveness in a managerial context emerges from its capacity to foster collaboration, strengthen commitment to solutions, and align executive teams. The study revealed that DT engagements encourage a collaborative environment in which executives contribute their expertise, leading to more comprehensive and innovative solutions.

Collaborative decision-making and alignment

The participatory nature of DT helps reconcile differing perspectives within the leadership team. Executives and leaders must be actively engaged in the strategic management challenge, as strategic decisions frequently exceed the cognitive capabilities of any single leader due to complexity and scale (Andersen, 2023).

> *"The joint definition of the solution transforms the willingness from goal orientation to embracing it as a personal agenda, to make it a success. Managers only commit to*

simple-to-understand and especially clear solutions." (Interviewee #2)

This collaborative approach strengthens alignment and ensures that strategies are co-created, thereby increasing the likelihood of successful implementation. This finding supports Lund Strøm et al. (2018), who argue that strategies are more likely to succeed when both the strategy and the corresponding high-level execution plan are defined collaboratively.

Commitment to outcomes

Managerial DT solutions are typically intangible, business architectural, or strategic funneling into feasible corporate programs for implementation. Although solution fidelity or the "prototype" can be important, the collaborative artifact-creation process often matters more for management teams. This process fosters commitment to and confidence in executing the solution concept, echoing findings from Liedtka (2017). Studies by Carlgren, Elmquist, and Rauth (2014) show that conceptualization and prototyping make ideas more graspable, functioning as a common language that supports decision-making. Communication and increasing commitment reduce organizational formalism through trust-building. However, this presumes the attendance of a critical number of managers and executives with domain knowledge and empowerment to make joint decisions.

"Low-fidelity 'prototypes' or management solutions are good enough if they carry the message and describe the idea to which executives and their delegates can commit." (Interviewee #2)

Challenges in implementing Design Thinking in management contexts

While DT has proven effective in managerial settings, several challenges arise, primarily due to time constraints, methodological peculiarities, and managerial behaviors.

Methodological challenges

Time constraints and iteration limitations Managerial DT engagements are typically short, often limited to one or two days due to executives' demanding schedules. Additionally, managers and executives may resist iteration once a consensus is reached, even if time permits. This mindset restricts iteration, a hallmark of traditional DT methods (Brown, 2008; Plattner et al., 2009). Because of the short- and single-cycle time constraints, correcting a misguided engagement is thus almost impossible. The lack of iteration must be compensated by the facilitator's comprehensive preparation, coaching experience, and industry or business domain knowledge. Facilitators need to ensure that the engagement is well-structured to maximize the limited time available.

"...does not really allow for iteration; the single cycle is important in doing it right. With managerial DT, you have only one shot." (Interviewee #3)

Dilemma of human-centeredness

Traditional DT relies on human-centric techniques such as Personas and Customer Journey Mapping (Johansson-Sköldberg et al., 2013; Micheli et al., 2019). However, in highly

abstract managerial challenges, these techniques may be ill-suited. Interviewees emphasized the importance of selecting techniques that align with the challenge's context and cautioned against forcing human-centric methods where they do not fit. This human-centeredness dilemma can affect both problem exploration and solution development.

> *"If a technique or a principle from DT is not fitting with the management challenge, don't force fitness. If a persona does not fit in a very high-level management workshop, then keep it out." (Interviewee #1)*

Managerial challenges

Impatience with problem exploration Executives often exhibit impatience during the problem exploration phase, preferring to move quickly to solution development while assuming they already possess the necessary insights. This impatience can limit the depth of understanding and affect outcome quality. Facilitators must manage this tendency by insisting on sufficient time for problem exploration, ensuring sufficient insights are collected before moving to solution development.

> *"Managers do not have the patience to bear for completion of problem space, assuming they have already understood the problem. Managers see limited value because the problem space does not generate the ultimate result in terms of a solution." (Interviewee #2)*

Influence of hierarchical dynamics

Strong personalities or hierarchical structures can dominate discussions, potentially undermining diverse perspectives and pushing their agendas. Confident DT facilitators handle these dynamics and maintain a balanced and collaborative environment. This aligns with findings from Carlgren et al. (2016), who emphasize the importance of facilitation skills in managing group dynamics during DT engagements.

> *"Confident DT engagement leads can handle impatient and strong personalities." (Interviewee #2)*

Maintaining engagement amidst competing priorities

Executives' busy schedules and competing priorities may lead to intermittent participation, disrupting the engagement flow and weakening team cohesion. Facilitators need to plan engagements carefully to mitigate such disruptions.

Success factors for managerial design thinking

The study identifies several key factors critical to DT's success in managerial contexts.

Effective preparation and facilitation

Due to the limited time and high stakes of managerial engagements, meticulous preparation is indispensable. Facilitators must have domain knowledge and experience to guide executives effectively. Experienced facilitators can manage strong personalities

and ensure that sufficient insights are collected in the problem space before moving to solution development.

> *"The engagement requires higher effort for preparation than with classic DT. The preparation ensures a smooth start; acceptance and confidence start in the first seconds when people meet."(Interviewee #3)*

Tailored selection of techniques

A business problem exploration activity is directly followed by its insight generation activity. Multiple dualistic activities can then feed into a synthesis framework that concludes the problem space, unveiling patterns, interdependencies, and aggregate insights. The careful orchestration of business techniques can significantly impact the generation of insights for business problem exploration. Executives prefer simple and effective business techniques, though they are open when the situation requires more complex techniques. For optimal business value, it is necessary to maintain a balance between the effort required for an activity and its eventual outcome. Pre-structured and template-based business techniques bolster confidence among participants because of their given structure; however, any rigid structure can limit unexpected new insights. The prudent selection and combination of business techniques can enhance engagement and outcomes.

> *"The dualism of divergent and convergent activities per problem and solution space is typical for both classic and managerial DT." (Interviewee #1)*

Balanced focus on problem and solution spaces

Allocating adequate time to both problem exploration and solution development is vital. Facilitators must ensure that executives avoid rushing through the problem space, which could compromise solution quality.

> *"A badly covered problem space will not result in good outcomes." (Interviewee #2).*

Commitment through collaborative creation

Engaging executives in the co-creation of solutions fosters ownership and commitment. The collaborative artifact-creation process reinforces confidence in executing the solution concept. While the tangible output of managerial DT engagement is important, achieving alignment among executives is often even more critical to success. This aligns with findings from Carlgren et al. (2014), who suggest that prototyping and conceptualization function as a common language and support decision-making.

> *"The joint definition of the solution transforms the willingness from goal orientation to embracing it as a personal agenda, to make it a success." (Interviewee #2)*

Adaptations and peculiarities in managerial contexts

Adapting DT for managerial contexts requires recognizing specific nuances and making targeted adjustments.

Expanded methodological span

Managerial DT engagements often have a more expansive methodological span than traditional DT methods. Additional preparation and implementation phases are often required to accommodate executives' needs and the engagement's objectives. This finding aligns with Laursen and Haase (2019), who note that emerging DT methods tend to be more expansive and incorporate extended processes.

> *"The engagement cycles are rather short, but the end-to-end method span is longer." (Interviewee #3)*

Preference for sequential methods

Executives often favor sequential represented DT methods over circular ones. Sequential methods offer a clear progression, aligning with executives' desire for structure and tangible progress. They allow executives to track the gradual progress of the engagement, which is particularly important given their limited availability.

> *"In managerial DT engagements, almost only sequential DT methods have been used." (Interviewee #3)*

iii) Acceptance of conceptual solutions

In managerial DT, outcomes are often intangible or conceptual, such as strategies, roadmaps, implementation plans, methods, new organizational structures, process models, value chains, business architectures, or innovative business models. These prove valuable to executives if they effectively convey the idea and are actionable. This suggests that the fidelity of the solution or "prototype" is less important than its clarity and the commitment it generates. Additionally, the term "prototyping" in DT methodology can confuse executives, as they often associate it with physical prototypes used in discrete or chemical industries.

Validating managerial DT solutions is difficult, as every solution is unique to the management challenge, corporate culture, and customer base. However, validating parts of the solution, through management consultancies, research institutes, best practices, minimum viable business solutions, forecasts, and simulation, remains feasible.

> *"The idea needs to be conceptualized. The concept from the workshop is the value of managerial DT." (Interviewee #3)*

Integration with management techniques

Managerial DT effectively merges DT principles with traditional management tools. The more abstract the management challenge, the broader the applicability of various techniques and tools. Conversely, the more precise the management challenge, the more specific the business techniques. Organizational interdependencies, including constraints, threats, pain points, and corporate needs synthesized in a structured framework, often represent typical management insights in the problem space.

The integration of DT-typical and management tools ensures that the DT process is aligned with executives' expectations and organizational practices, enhancing acceptance and effectiveness.

> *In 90 percent of the managerial DT engagements, the coach has to live with the existing knowledge in the room for data and fact collection (Interviewee #2).*

Table 3 presents the categorized conclusion from the research insights illustrating the characteristics and challenges of managerial DT.

Accordingly, managerial DT can be defined as a systematic management method for addressing holistic, systemic, or strategic management challenges within an organization. Engagement participants include stakeholders, managers, and executives who collaborate to design the management solutions while adhering to the principles of DT. It begins with a mutual problem comprehension of the business challenge before commencing the business design process. Managerial DT integrates descriptive and prescriptive techniques from classic DT and management frameworks. The outcome of a managerial DT engagement can be intangible or conceptual. It is an artifact for decision-making and commitment, typically leading to business transformations and reinvention.

Discussion

Most of the existing DT research in business examines its organization-wide adoption to increase innovativeness and foster an innovation culture (Brown, 2008; Carlgren et al., 2014, 2016; Dunne & Martin, 2006; Hassi & Laakso, 2011a, 2011b; Liedtka, 2014; Liedtka & Ogilvie, 2011; Rhinow & Meinel, 2014; Schmiedgen et al., 2016). In contrast, this research focuses on DT as a practice that management uses to solve its strategic challenges. It investigates the peculiarities of managerial DT and demonstrates its suitability under the balanced consideration of multiple engagement factors and adherence to DT principles. The emphasis is on the systemic view, shorter time availability, and the primary audience of executive participants who address strategic business challenges. The research also highlights the need for a reliable managerial DT method and underscores DT's potential as a creative leadership method.

The research findings demonstrate a robust correlation between design challenge depth, engagement time, iterability, and solutions. Notably, the lack of iteration can be attributed to the time constraints and managers' progression-oriented mentality. While the resulting management solutions may lack the granular detail characteristic of high-fidelity product prototypes emerging from extended, human-centric DT processes, they prove suitable for strategic decision-making and the initiation of transformative corporate programs.

Prior studies confirm that DT helps overcome the strategy-to-execution gap by reducing uncertainties about implementation through collaborative co-creation. Implementation risks are mitigated through the involvement and buy-in of stakeholders, decision-makers, and executors, who are considered 'users of the strategy' (Elsbach & Stigliani, 2018; Knight et al., 2020; Lund Strøm et al., 2018). Consequently, DT's range of addressing management problems is broad, extending from strategic to the operational levels, encompassing from business transformation definitions to their associated

Table 3 Peculiarities in managerial design thinking

Methodological insights	Managerial implications	Outcome value
Business design challenges are focused on solving management problems sustainably and less focused on (management) innovation perspective	Management problems are focused on holistic organizational issues instead of single business personas. The problem types are rather systemic, abstract, or strategic	The business artifacts and solution concepts bridge strategy design and implementation
The engagement cycles are rather short, but the end-to-end method span is longer	The engagements are collaborative, democratic, and complementary on expertise	Business artifacts are low-fidelity and conceptual, but sufficiently suitable to get commitments and embark
Executives and management participants prefer simple (business) techniques, unless the value of complex techniques is significantly high	Managers are impatient with business problem comprehension. They are progression-driven demanding acceleration toward solving	The management solution concepts serve for discussion, decision, initiation and tracking of corporate programs
Managerial DT is highly interoperable with other modern management methods	Generative management collaboration leads to faster consensus-building and stronger commitment	Examples of solution concepts are strategic roadmaps, roles and responsibility distribution, corporate initiative, or implementation-oriented

change management processes (Silva & Zancul, 2023; Baran, 2017; Bühring & Liedtka, 2018; You, 2022).

Design Thinking often uncovers novel or additional data points that can be incorporated into strategizing by combining deductive, inductive, and abductive reasoning. The multi-stakeholder perspectives of DT also prove beneficial in developing unorthodox management solutions. Research indicates that DT is particularly effective in creating strategic opportunities because of its ability to apply deductive, inductive, and, most importantly, abductive reasoning. Abductive reasoning significantly impacts innovative business model generation (Garbuio et al., 2018; Guldmann et al., 2019; Micheli et al., 2019; You, 2022).

This study yielded several unexpected findings. In contrast to classic DT engagements, the methodological span of managerial DT is often more expansive. This expansion stems from the need for enhanced alignment and engagement preparation at the outset, coupled with additional implementation-related activities that lead to corporate program initiation. The expansion is necessary to efficiently and effectively conduct a managerial engagement, where time and the availability of executive audiences are critical factors. A recent study by Laursen and Haase (2019) indicated that emerging DT methods are generally more expansive with a more extended DT method 'process', which aligns with the findings of this research.

It may be assumed that generic or abstract challenges are more easily addressed through DT engagements. However, interestingly, the application of DT theory is often clearer and can be applied more rigorously in user-centric engagements that span a longer duration than the compressed timelines characteristic of managerial DT engagements. In traditional DT processes, the inclusion and integration of the end-user, or the affected persona, are inherently integral to the methodology. In such instances, the tangible product or service at the heart of the innovation process acts as an intermediary between the design challenge and the individuals it affects.

Despite executives' familiarity with high-pressure environments, the alternation between divergent and convergent thinking in Design Thinking sessions can induce mental fatigue due to continuous ideation and decision-making cycles. Some managers' limited grounding in business theory or management education may lead to underestimating the complexities of managerial design challenges. To address this, integrating implicit educational components through the involvement of domain experts, such as professors specializing in management, can prove beneficial. Reframing strategic management challenges until executives embrace them as personal or coherent team challenges is similarly advantageous.

The lengthy problem statements or managerial design challenges inherent in Design Thinking methodology, which exhaustively describe problems, scope, context, and constraints in single sentences, often appear unconventional to executives accustomed to condensed information formats. Occasionally, the democratic culture of DT also conflicts with traditional hierarchical decision-making structures and delegation practices.

Because of their demanding schedules, executives may intermittently disengage from managerial DT engagement to address urgent matters or attend pivotal meetings, returning with a disconnect that can impede the progress of other team members or create confusion regarding prior decisions. Leaders' and executives' commitments often

materialize more quickly when proposed solutions align with their managerial agendas, confidence in peers is high, and the overall solution or strategy is convincing. This commitment is especially evident when executives are directly involved in ideation or strategic conceptualization processes.

Our findings also reveal that executives prefer sequential representations of DT methods over circular ones. They value the ability to track an engagement's progress and favor a phased approach with a tangible target, akin to a "finishing line". Additionally, we found that consistent adherence to DT principles supersedes rigid methodological application in importance.

Human-centric techniques such as Persona, Customer Journey Maps, and Point-of-View prove less applicable to strategic and abstract management challenges. In these cases, the abstraction of management challenges demands business-specific techniques to empathize on an organizational level. Other studies similarly indicate potential incongruities and suggest that DT should extend beyond human-centricity (Carlgren et al., 2014; Laursen & Haase, 2019). The dualism of divergent and convergent activities across problem and solution space remains characteristic of both classic and managerial DT. This dualism ensures sufficient exploration and insight creation in the problem space as well as robust business idea generation and conceptualization in the solution space.

These findings and observations do not necessarily imply the exclusion of circular representations or human-centric techniques in managerial DT. With its visual emphasis on iteration, circular DT representations can challenge executives and reinvigorate abductive thinking and creativity for management solutions. Similarly, judiciously applied human-centric techniques can ensure that strategic decisions remain grounded in stakeholder needs, particularly regarding customers and employees. While managerial DT emphasizes short-cycling and time constraints for executives, one must recognize the value of longer-cycle engagements, which may yield a deeper understanding of the management challenges and, therefore, build more refined solutions and strategies.

This research was conducted within the specific context of managerial DT, which may limit the generalizability of the findings. The context of DT application on managerial challenges and its target executive audience is distinctive, and these insights may not apply to other generic design challenge types. The context and associated management challenges are typically characteristic of large, traditional enterprises in predominantly mature markets. The experiences and applicability of managerial DT may also vary for companies of different sizes or organizations within emerging markets. Additionally, this study was carried out with DT practitioners primarily working with leadership from investment-affine and largely market-leading companies. Such individuals may inherently hold a more open and forward-thinking entrepreneurial perspective than other managers, who may be more inclined toward established practices. As a result, the generalizability of this study to other large enterprises may be constrained by leadership mindset, capability, and economic decision-power. Moreover, the study's insights might not extend to mixed-team engagements of generalists and specialists.

This study focused on engagements involving constructive group sizes of four to eight participants. Concerns arise regarding whether smaller groups can yield high-quality outcomes. A primary benefit of managerial DT is the participants' commitment to outcomes and implementation. Consequently, having fewer leadership

participants may reduce the likelihood of successful strategy execution. There are also concerns about engagements with more than eight participants, as extended discussions may undermine the efficiency of constructive decision-making and dilute the depth of commitment.

Qualitative studies yield highly detailed insights but involve a trade-off in scope. Interviews, as a qualitative research method, typically rely on smaller sample sizes than surveys, potentially imposing more substantial limitations on the generalizability of findings.

Such a narrow selection of experienced DT experts and facilitators may introduce biases rooted in their pragmatic experiences with managerial DT. The potential for conflicts of interest between methodological adherence and client expectations has also been acknowledged. As a result, the findings from the interview-based qualitative study may be inherently subjective. Interviewees may also be less inclined to discuss unsuccessful DT applications, leading to an overrepresentation of successful cases. Moreover, selecting DT experts who have a wide range of experiences may not accurately represent the average experience levels, as it could favor those with more frequent positive outcomes. Familiarity with specific DT tools and techniques was not a selection criterion, potentially introducing further variability.

Several avenues for external validation and expanded research emerge from this study. The application of managerial design thinking could extend beyond organizational boundaries to involve strategic business partners and stakeholders, potentially reinforcing individual company strategies across broader business networks. Given its compact and strategy-oriented format, managerial DT may also be transferable to other strategic challenges beyond the purview of management, such as socio-economic or macro-economic concerns. Moreover, it is plausible that middle management and other organizational teams could leverage managerial DT's guided and structured approach for 'strategizing' and optimizing within their ecosystem, suggesting potential synergies across all organizational levels.

Structured and semi-structured interviews require neutral facilitation and a deep understanding of the subject, as responses can be highly detailed. The nature of qualitative studies allows for potential interviewer influence on responses and variability in interpretation. Additionally, coding bias can affect consistency and reliability.

The qualitative approach using interviews proved appropriate and purposeful. Although the contextual specificity does limit the generalizability of the findings, it simultaneously provides highly detailed and rich insights within this narrower scope of managerial DT. The results enabled an in-depth exploration with highly experienced DT experts addressing management challenges for a concrete leadership audience, while also highlighting specific improvements in method and applicability.

The foregoing limitations could affect the study's validity and reliability if the research method is not rigorously and objectively applied. Maintaining a clear research scope and structured interview design is essential for preserving the study's validity. The study's potential replicability bolsters reliability. The interview avoided purely open-ended and unstructured questions. Reliability was further reinforced by introducing broader thematic elements and coding categories to which the coded data could be systematically assigned.

Conclusion

This study examined the applicability and effectiveness of DT as a management method for addressing complex strategic challenges in organizations. The findings reveal that DT serves as a transformative tool in strategic management to address 'wicked' management problems. Specifically, the study shows that DT facilitates collaboration among executives, enhances decision-making and commitment, and aligns management solutions with systemic organizational needs. Significantly, the research demonstrates that DT, while traditionally associated with design and innovation domains, effectively extends into high-level management contexts, addressing limitations inherent in conventional management methodologies.

Design Science Research (DSR) has been employed as both a theoretical and practical framework, bridging the gap between theory and practice through predominantly qualitative research methods. DSR facilitated an in-depth exploration of Design Thinking as a managerial tool, positioning it as a research artifact within real-world contexts. This approach enabled a comprehensive examination of DT's applicability and efficacy in addressing complex strategic management challenges. The artifact-centric nature of DSR proved instrumental for this study, allowing for the systematic identification, application, and critical evaluation of DT within managerial settings. It has also ensured that the research remained grounded in practical realities. Furthermore, the rigorous methodological structure of DSR substantially contributed to enhancing the reliability and validity of the study's findings.

By employing DSR, this study has closed notable gaps in existing research, effectively linking its findings to established theories while also contributing new research insights. DSR provided a robust framework for systematically exploring and evaluating DT's role and impact in strategic management, particularly in solving complex, 'wicked' problems faced by executives in large enterprises. This research contributes to the academic discourse on DT and its strategic applications, offering practical insights for organizations seeking to implement DT as a part of their strategic management toolkit.

This study contributes to managerial DT research by providing further methodological insights, assessing compatibility, and demonstrating added value. The full systems view of Design Thinking remained underexplored and was open for further research (Shapira et al., 2017). This research addresses that gap by offering specific explanations to DT's compatibility with the full systems view and by examining the potential mismatch of applied tools and techniques. Compared to the widely researched end-user or individualized perspective, these findings contribute to the managerial perspective of DT (Bühring & Liedtka, 2018; Elsbach & Stigliani, 2018; Liedtka & Kaplan, 2019; Knight et al., 2020). This study aligns with DT's management discourse, where DT is regarded as a tool for strategy formulation, business innovation, and strategic decision-making, and is also promoted as a cognitive capability for managers to solve complex business challenges (Baran, 2017; Bühring & Liedtka, 2018; Matthews & Wrigley, 2017).

Our research advances both cognitive and practice theories of managerial DT, expanding the current knowledge base (Knight et al., 2020). The theories of Martin (2006) and Liedtka (2014) laid the foundation for integrating the cognitive and practice perspectives of Design Thinking into strategic management, and our findings build on their theory of abductive strategies for management. Other studies have confirmed that DT is a valuable

method for management innovation, facilitating the generative and prescriptive solving of strategic business problems. Managerial DT exhibits enhanced decision-making capabilities through its generative and creative approach (Baran, 2017; Elsbach & Stigliani, 2018; Garbuio et al., 2018; Knight et al., 2020). Since strategizing involves a practice perspective, our research aligns with the paradigm of 'strategy as a practice' using DT as a source (Brenner et al., 2016; Çeviker-Çınar et al., 2017; Kremel & Wetter Edman, 2019). Knight et al. (2020) refer to this as "…how strategists can integrate Design Thinking into strategizing in the moment."

The study yields several practical recommendations for implementing Design Thinking in managerial contexts. We advocate for a tailored Design Thinking methodology that specifically addresses systemic and abstract managerial challenges while accommodating the unique needs of an executive audience, namely, time constraints and the requirement for actionable solutions. Practitioners and DT experts applying managerial DT must be mindful of the engagement cycles and the overall method span. While engagement cycles in DT are relatively short- and single-cycle, the methodological span typically exceeds that of traditional DT methods and involves additional phases. This demands thorough planning and preparation to ensure effective DT application. A significant practical implication is the wise orchestration of value-adding tools and business techniques for each phase. These should purposefully focus on the challenge complexity, solution orientation, and the audience's ability to understand and utilize these techniques effectively.

Despite the time constraints typically associated with executive engagements, allocating sufficient time for exploring the problem space before transitioning to the solution space remains critical. The DT process should consistently foster an encouraging, thought-provoking, and challenging environment across multiple management levels. Such an atmosphere is pivotal for cultivating creative and effective solutions.

Design Thinking facilitators must demonstrate competence in managing power dynamics among managers and executives while ensuring a democratic engagement environment. Early in the engagement, it is essential to clarify that outcomes are largely contingent on co-creation and commitment from all involved parties. While these outcomes might be conceptual or low fidelity, their primary role is to facilitate decision-making and consensus-building, leading to the approval and implementation of management solutions.

A DT expert or facilitator with specialized subject or industry knowledge should be integrated into the process. Their expertise is invaluable in challenging teams to produce superior outcomes and effectively steering the overall engagement. Such facilitators play a critical role in navigating the unique dynamics of managerial DT, ensuring the methodology is applied effectively and yielding meaningful results.

Future research opportunities exist in further investigating the direct relationship between measurable corporate success and decision-making with managerial DT. Additional studies may also be required to understand the nuances and complexities of managerial design challenges in the DT context, particularly how their definitions and specificity influence the efficiency and effectiveness of managerial DT. Future research could also investigate how a management-centric DT approach might enhance the adoption of design-driven leadership and management innovation. Furthermore, comparing

the performance of other management methods with managerial DT could yield valuable insights into the relative benefits of different approaches. Further inquiry might explore how DT can catalyze broader organizational transformation. Although a linear DT model was chosen in this research for didactic purposes and in response to executive preferences, future research could elaborate on circular method representation to assess solution quality and engagement cycle. Investigating patterns and correlations between tool or technique appropriateness, different types of strategic challenges, and method compliance would also be worthwhile. While our findings indicate dilemmas with human-centric tools and techniques, it would be valuable to investigate the highest possible level of management challenges that add more value using primarily human-centric tools and techniques. A reversed perspective, focusing on unsuccessful management engagements with DT, could shed light on the criteria that corrupt engagements. Finally, similar studies could extend this managerial DT research to differently sized companies.

Further research opportunities exist in developing a comprehensive evaluation framework for applying managerial DT. Such a framework could cover several key dimensions, including problem identification, congruence with overarching corporate strategy, cultural readiness for managerial DT adoption in management, resource availability, methodological coherence with existing management practices, and estimation of strategic impact.

This research on applying DT in a managerial context for an executive audience has yielded valuable insights. The brevity of the engagement, typically one or two days, offers advantages by assembling executives quickly. A sequential and intuitive DT method is more readily accepted by managers, providing a sense of control. The well-definition and granularity of the management challenge significantly influence the engagement's overall success. Typically, any managerial DT engagement can be planned as a single- and full-cycle without real iteration possibilities. Consequently, the selection of business techniques should be deliberate to ensure a harmonious start and sustained executive collaboration. It is equally important to involve a complementary, heterogeneous mix of executives, managers, and external thought leaders. Intrinsically motivated executives with complementary business expertise not only commit to solutions but also possess the authority to realize them. The value orientation and generative nature of managerial DT transcend traditional responsibility lines, fostering peer empathy and cross-domain learning among executives. DT's focus on "doing" can help executives become generative decision-makers, capable of bridging the strategy-to-execution gap in large enterprises. The generative element of managerial DT solutions also opens up new business opportunities and perspectives, ushering in new avenues for management innovation.

The development of a time-sensitive, sequential managerial DT approach using a simplified method language may increase leadership acceptance. However, this study demonstrates that the method represents only one critical component of the overall approach. The managerial DT approach requires strict adherence to DT principles, as well as close attention to techniques, facilitation, preparation, timing, and participatory engagement of executives and managers.

Abbreviations

DT Design Thinking
DSR Design Science Research

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Declarations

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